

Project Synopsis: Virtual Herbal Garden

Project Title: Virtual Herbal Garden

Team Members:

NNM23CS083 Himanshu Poojary

NNM23CS110 Mithesh R Shetty

Objective:

To develop a web-based application called 'Virtual Herbal Garden' that showcases medicinal plants used in AYUSH (Ayurveda, Yoga & Naturopathy, Unani, Siddha, and Homeopathy). The project aims to create an interactive platform for learning about the uses, benefits, and characteristics of medicinal plants.

Scope:

- Display detailed information on various medicinal plants.
- Implement interactive 3D models using Three.js.
- Incorporate a searchable and filterable database of plants.
- Provide multimedia integration for an enhanced learning experience.
- Offer a virtual tour feature for guided exploration.

Technology Stack:

- Frontend: Next.js for building a fast, responsive, and SEO-friendly interface.
- Backend: Next.js API routes.
- Database: MySQL or PostgreSQL for structured data management.
- 3D Models: Three.js for interactive visualizations.

Features:

1. Plant Information Database: Comprehensive data on medicinal plants, including their scientific names, uses, benefits, and cultivation methods.
2. Interactive 3D Models: Users can explore 3D models of plants for better understanding.
3. Search and Filter Options: Efficient search to find plants based on health benefits, regions, and types.
4. Multimedia Integration: Images and videos to support textual content.
5. Virtual Tour: An immersive, guided tour of a virtual herbal garden.

Database Schema:

- Plants Table: PlantID (PK), Name, ScientificName, Description, Uses, Benefits, Region, ImageURL, VideoURL
- Categories Table: CategoryID (PK), CategoryName, Description
- PlantCategory Table: ID (PK), PlantID (FK), CategoryID (FK)
- Users Table: UserID (PK), Name, Email, Password, UserType (Admin/User)

Future Scope:

- Integration of AI/ML for plant identification and health-based recommendations.
- Expansion to include a broader range of medicinal plants.

Conclusion:

The Virtual Herbal Garden serves as an educational and interactive platform to spread awareness about the medicinal value of plants recognized in AYUSH. By integrating technology and traditional knowledge, it aims to become a valuable resource for students, researchers, and herbal enthusiasts.